INCH-POUND

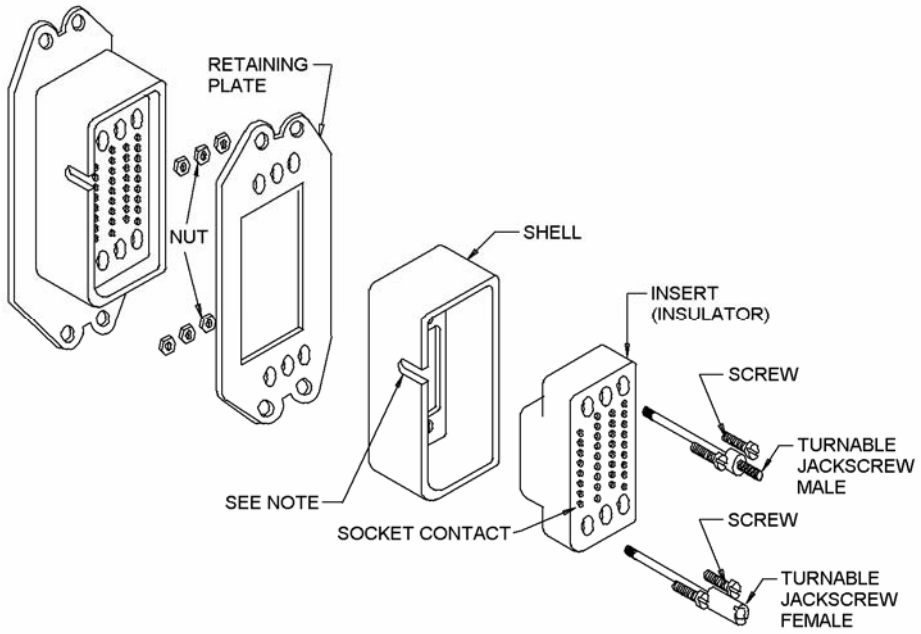
MIL-DTL-28748/4F
9 September 2004
SUPERSEDING
MIL-DTL-28748/4E
11 May 1999

DETAIL SPECIFICATION SHEET

CONNECTOR, ELECTRICAL, RECTANGULAR, RACK AND PANEL, CRIMP TYPE REMOVABLE SOCKET CONTACTS, SIZES 16 AND 20

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28748.



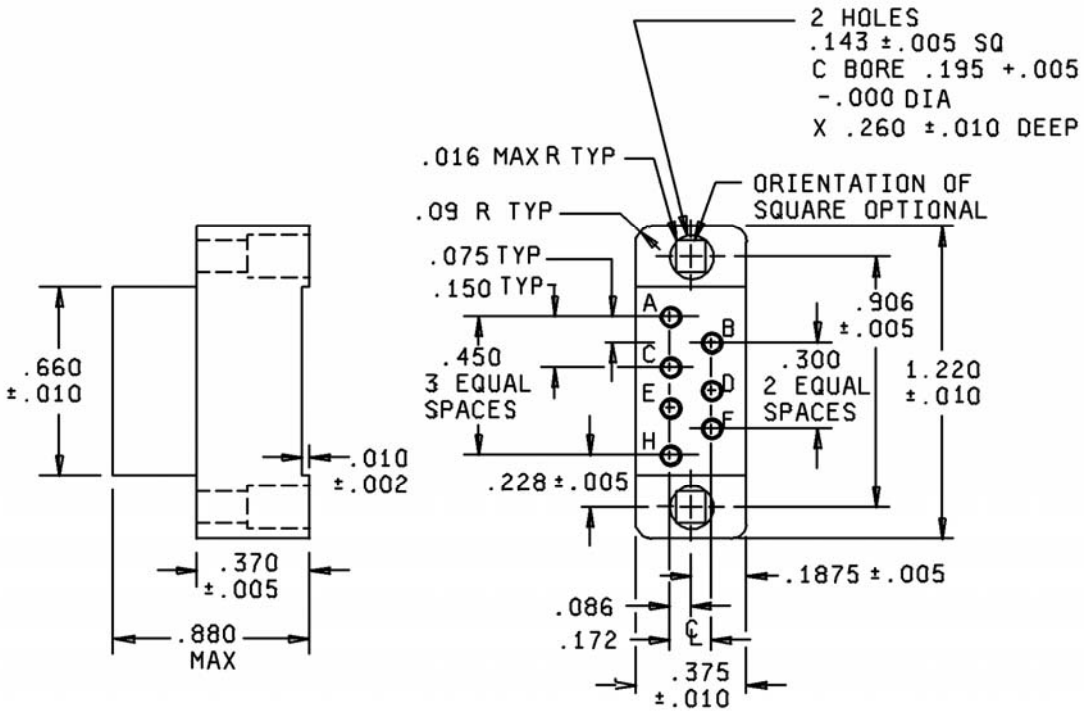
Connector

NOTE: The polarization feature (groove on the receptacle shell, pin on the plug shell) shall be adjacent to the side closest to the number 1 or A contact position.

Configuration A

FIGURE 1. Receptacle connector with retaining plate, receptacle shell, and turnable jackscrews.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.



Inches	mm	Inches	mm	Inches	mm
.002	0.05	.143	3.63	.300	7.62
.005	0.13	.150	3.81	.370	9.40
.010	0.25	.172	4.37	.375	9.53
.016	0.41	.1875	4.76	.450	11.43
.075	1.91	.195	4.95	.660	16.76
.086	2.18	.228	5.79	.880	22.35
.09	2.29	.260	6.60	.906	23.01
				1.220	30.99

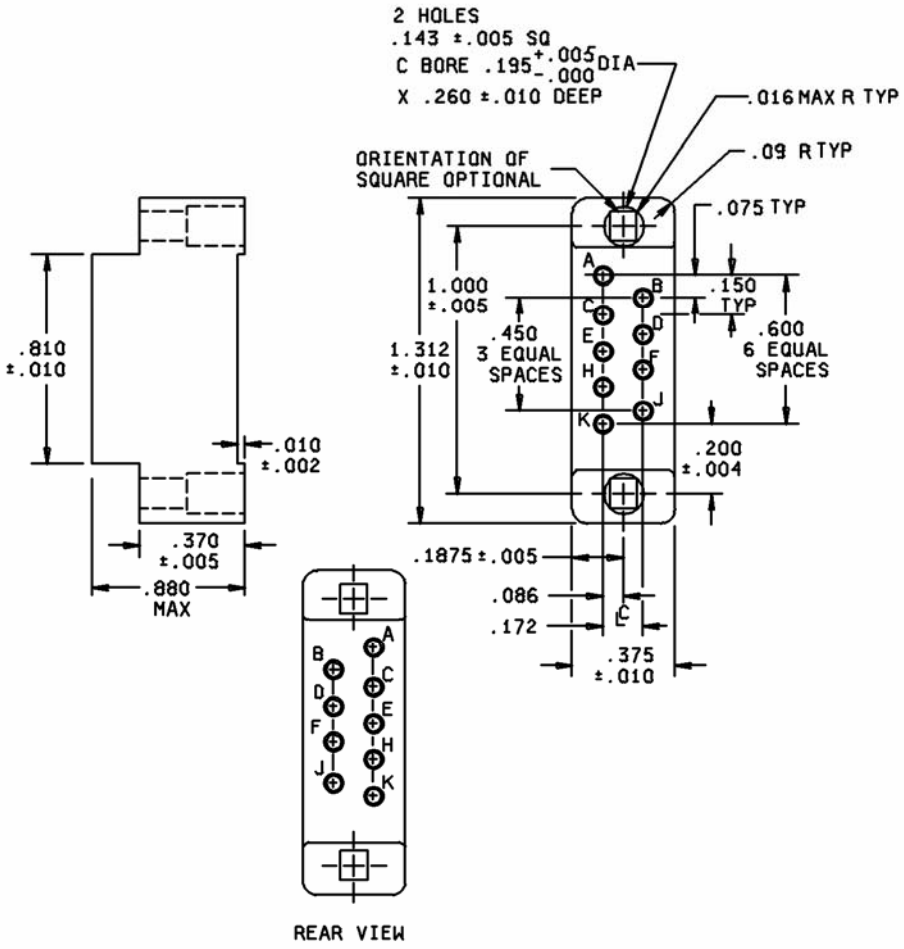
The maximum insertion and withdrawal force (connector assembly) shall be 8.75 lbf (38.92 N) pounds.

Insert with 7 contacts

Insert designator A

FIGURE 2. Insert arrangements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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Inches	mm	Inches	mm	Inches	mm
.002	0.05	.143	3.63	.375	9.53
.004	0.10	.150	3.81	.450	11.43
.005	0.13	.172	4.37	.600	15.24
.010	0.25	.1875	4.76	.810	20.57
.016	0.41	.195	4.95	.856	21.74
.075	1.95	.200	5.08	.880	22.35
.086	2.00	.260	6.60	1.000	25.40
.09	2.60	.370	9.40	1.312	33.32

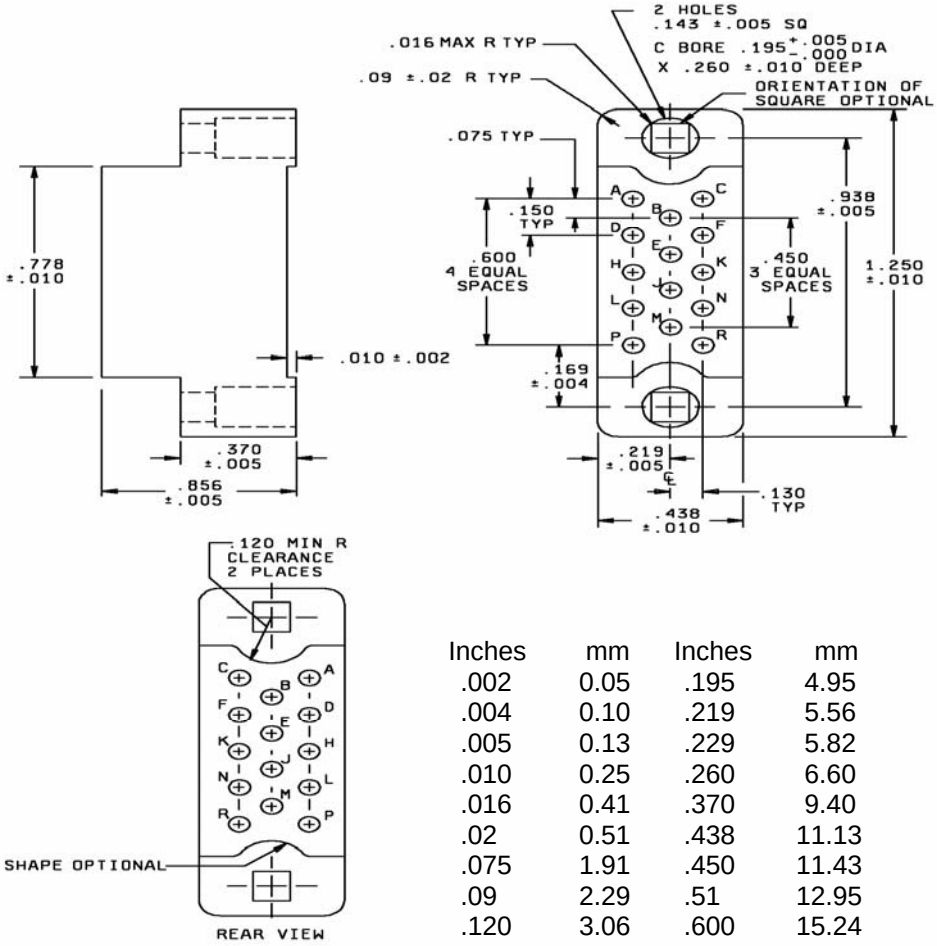
The maximum insertion and withdrawal force (connector assembly) shall be 11.25 lbf (50 N) pounds.

Insert with 9 contacts

Insert designator B

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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Inches	mm	Inches	mm
.002	0.05	.195	4.95
.004	0.10	.219	5.56
.005	0.13	.229	5.82
.010	0.25	.260	6.60
.016	0.41	.370	9.40
.02	0.51	.438	11.13
.075	1.91	.450	11.43
.09	2.29	.51	12.95
.120	3.06	.600	15.24
.130	3.30	.778	19.76
.143	3.63	.856	21.74
.150	3.81	.938	23.83
.169	4.29	1.250	31.75

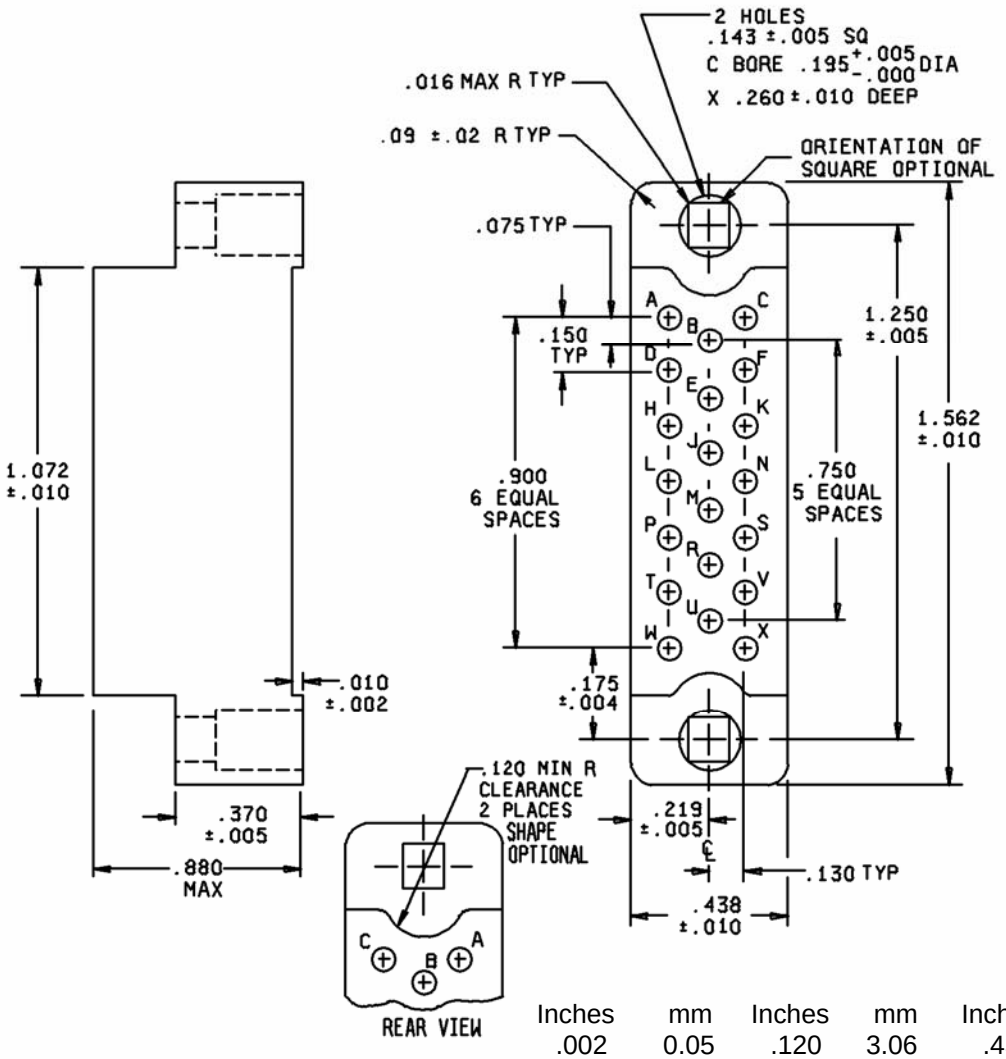
The maximum insertion and withdrawal force (connector assembly) shall be 17.5 lbf (77.84 N).

Insert with 14 contacts

Insert designator C

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.



Inches	mm	Inches	mm	Inches	mm
.002	0.05	.120	3.06	.438	11.13
.004	0.10	.130	3.30	.51	12.95
.005	0.13	.143	3.63	.750	19.05
.010	0.25	.150	3.81	.880	22.35
.016	0.41	.175	4.45	.900	22.86
.02	0.51	.195	4.95	1.072	27.23
.075	1.91	.219	5.56	1.250	31.75
.09	2.29	.260	6.60	1.562	39.67
		.370	9.40	2.290	58.17

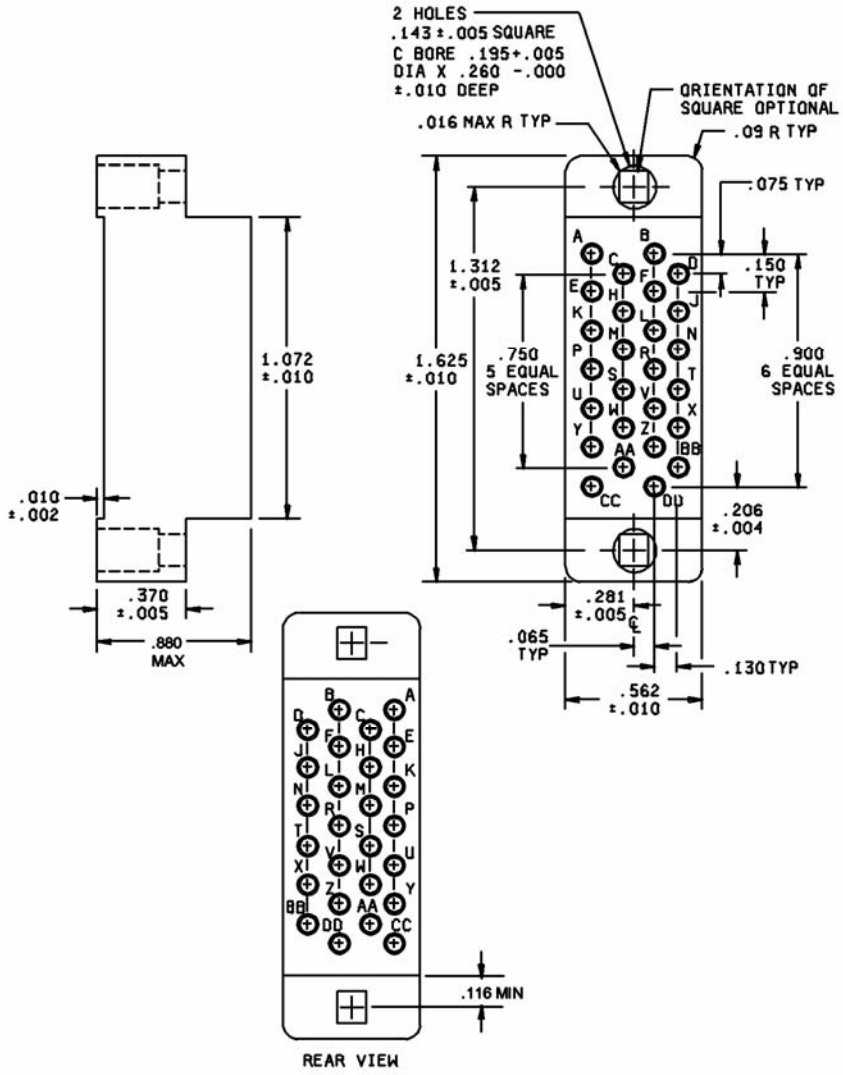
The maximum insertion and withdrawal force (connector assembly) shall be 25 lbf (111.21 N).

Insert with 20 contacts

Insert designator D

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
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Inches	mm
.002	0.05
.004	0.10
.005	0.13
.010	0.25
.016	0.41
.065	1.65
.075	1.91
.09	2.29
.116	2.95
.130	3.30
.143	3.63
.150	3.81
.195	4.95
.206	5.23
.260	6.60
.281	7.14
.370	9.40
.562	14.27
.750	19.05
.856	21.74
.880	22.35
.900	22.86
1.072	27.23
1.312	33.32
1.625	41.28

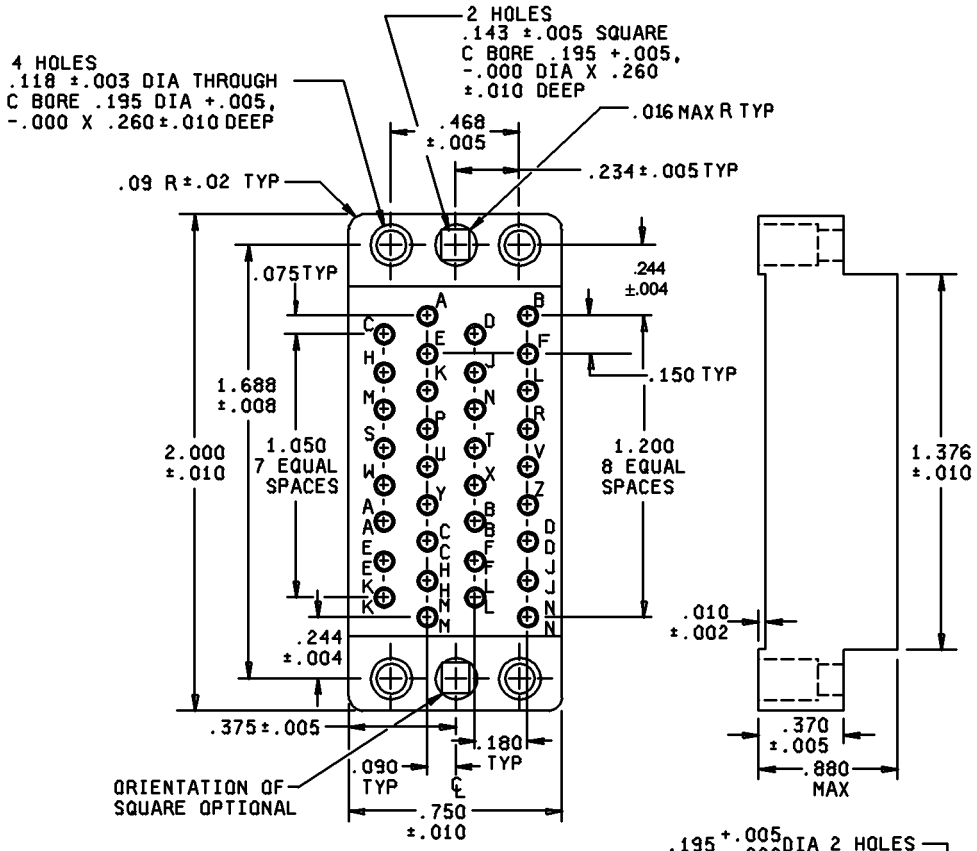
The maximum insertion and withdrawal force (connector assembly) shall be 32.5 lbf (144.57 N).

Insert with 26 contacts

Insert designator E

FIGURE 2. Insert arrangements - Continued.

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CENTER THROUGH
HOLE CONSTRUCTION
CONFIGURATION M

Inches	mm	Inches	mm	Inches	mm
.002	0.05	.118	3.00	.511	12.98
.003	0.08	.143	3.63	.750	19.05
.004	0.10	.150	3.81	1.050	26.67
.005	0.13	.180	4.57	1.200	30.48
.008	0.20	.195	4.95	1.350	34.29
.09	2.29	.234	5.94	1.376	34.95
.010	0.25	.244	6.20	1.500	38.10
.016	0.41	.250	6.35	1.672	42.47
.075	1.91	.260	6.60	1.688	42.88
.090	2.29	.370	9.40	2.000	50.80
.110	2.79	.375	9.53	2.290	58.17
		.468	11.68	2.312	58.72

Configuration F - Suitable for all applications, except for one-piece jackscrews.
Configuration M - For long jackscrews and shield only (through hole alternate construction).

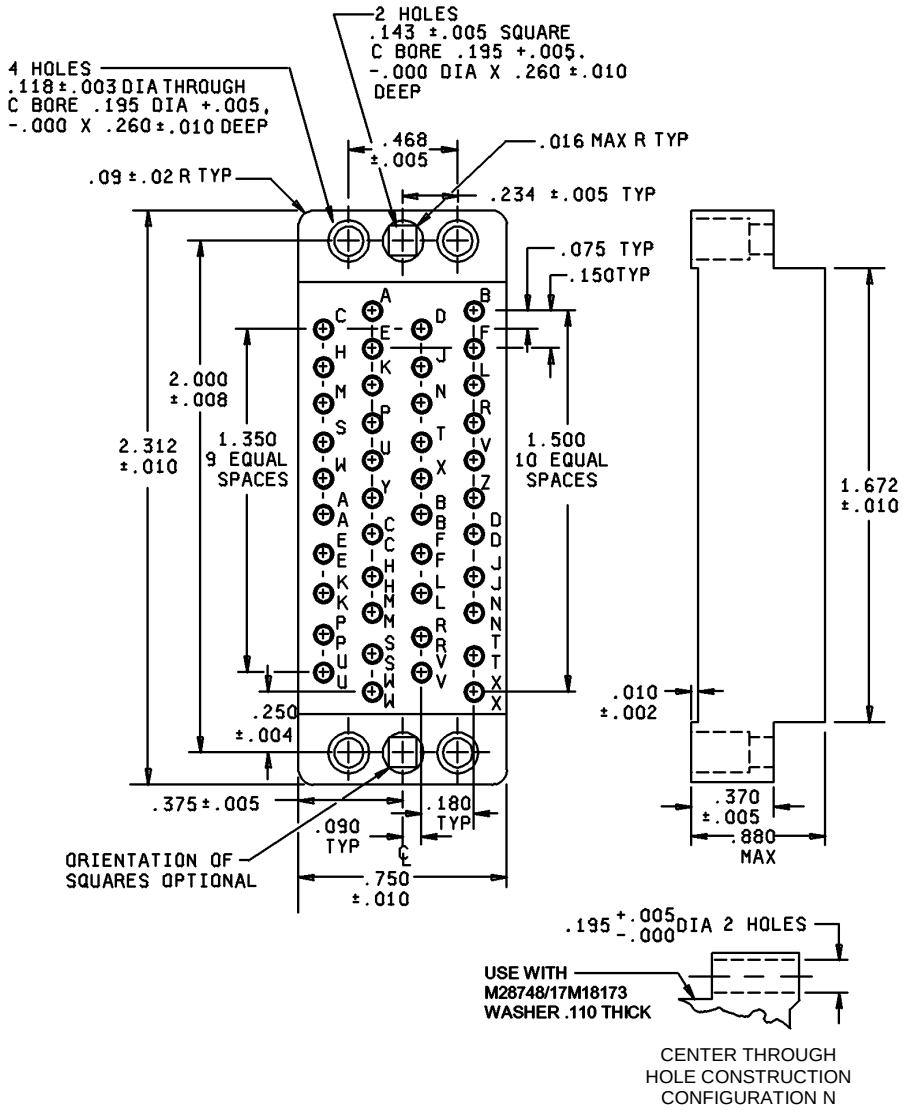
The maximum insertion and withdrawal force (connector assembly) shall be 42.5 lbf (189.05 N).

Insert with 34 contacts

Insert designator F and M

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.



Inches	mm
.002	0.05
.003	0.08
.004	0.10
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.02	0.51
.075	1.91
.09	2.3
.090	2.29
.110	2.79
.118	3.00
.143	3.63
.150	3.81
.180	4.57
.195	4.95
.234	5.94
.250	6.35
.260	6.60
.370	9.40
.375	9.53
.468	11.13
.51	12.95
.750	19.05
.856	21.74
.880	22.35
1.350	34.29
1.500	38.10
1.672	42.47
2.000	50.80
2.290	58.17
2.312	58.72

Configuration G - Suitable for all applications, except one-piece jackscrews.
Configuration N - For long jackscrews and shield only (through hole alternate construction).

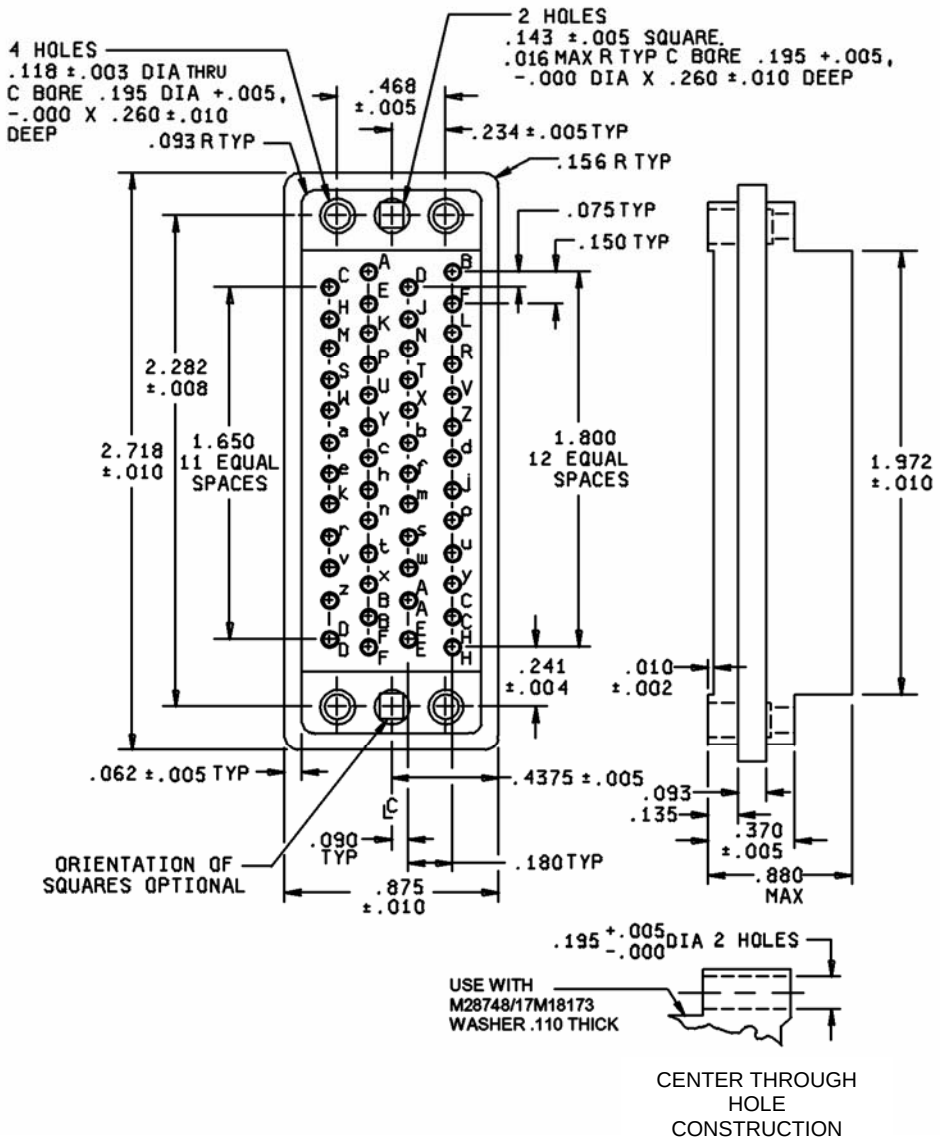
The maximum insertion and withdrawal force (connector assembly) shall be 52.5 lbf (233.53 N).

Insert with 42 contacts

Insert designators G and N

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.



CENTER THROUGH HOLE CONSTRUCTION

Inches	mm
.002	0.05
.003	0.08
.004	0.10
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.062	1.57
.075	1.91
.090	2.29
.093	2.36
.110	2.79
.118	3.00
.135	3.43
.143	3.63
.150	3.81
.156	3.96
.180	4.57
.195	4.95
.234	5.94
.241	6.12
.260	6.60
.370	9.40
.4375	11.11
.468	11.13
.856	21.74
.875	22.23
.880	22.35
1.650	41.91
1.800	45.72
1.972	50.09
2.000	50.80
2.282	57.96
2.718	69.04

Configuration H - Suitable for all applications, except one-piece jackscrews.
Configuration P - For long jackscrews and shield only (through hole alternate construction).

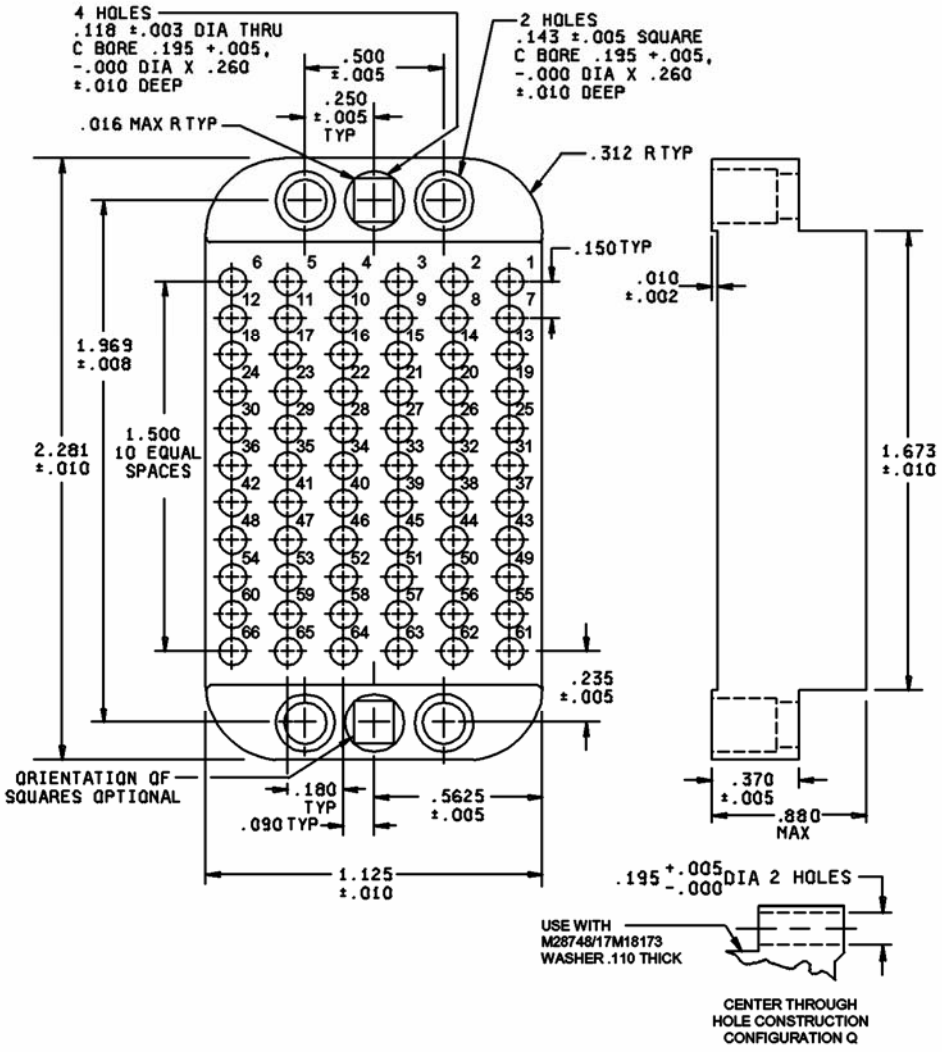
The maximum insertion and withdrawal force (connector assembly) shall be 62.5 lbf (278.01 N).

Insert with 50 contacts

Insert designators H and P

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.



Inches	mm
.002	0.05
.003	0.08
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.090	2.29
.110	2.79
.118	3.00
.143	3.63
.150	3.81
.180	4.57
.195	4.95
.235	5.97
.250	6.35
.260	6.60
.312	7.92
.370	9.40
.500	12.70
.5625	14.288
.880	22.35
1.125	28.58
1.500	38.10
1.673	42.49
1.969	50.01
2.281	57.94

Configuration J - Suitable for all applications, except for one-piece jackscrews.
Configuration Q - For long jackscrews and shield only (through hole alternate construction).

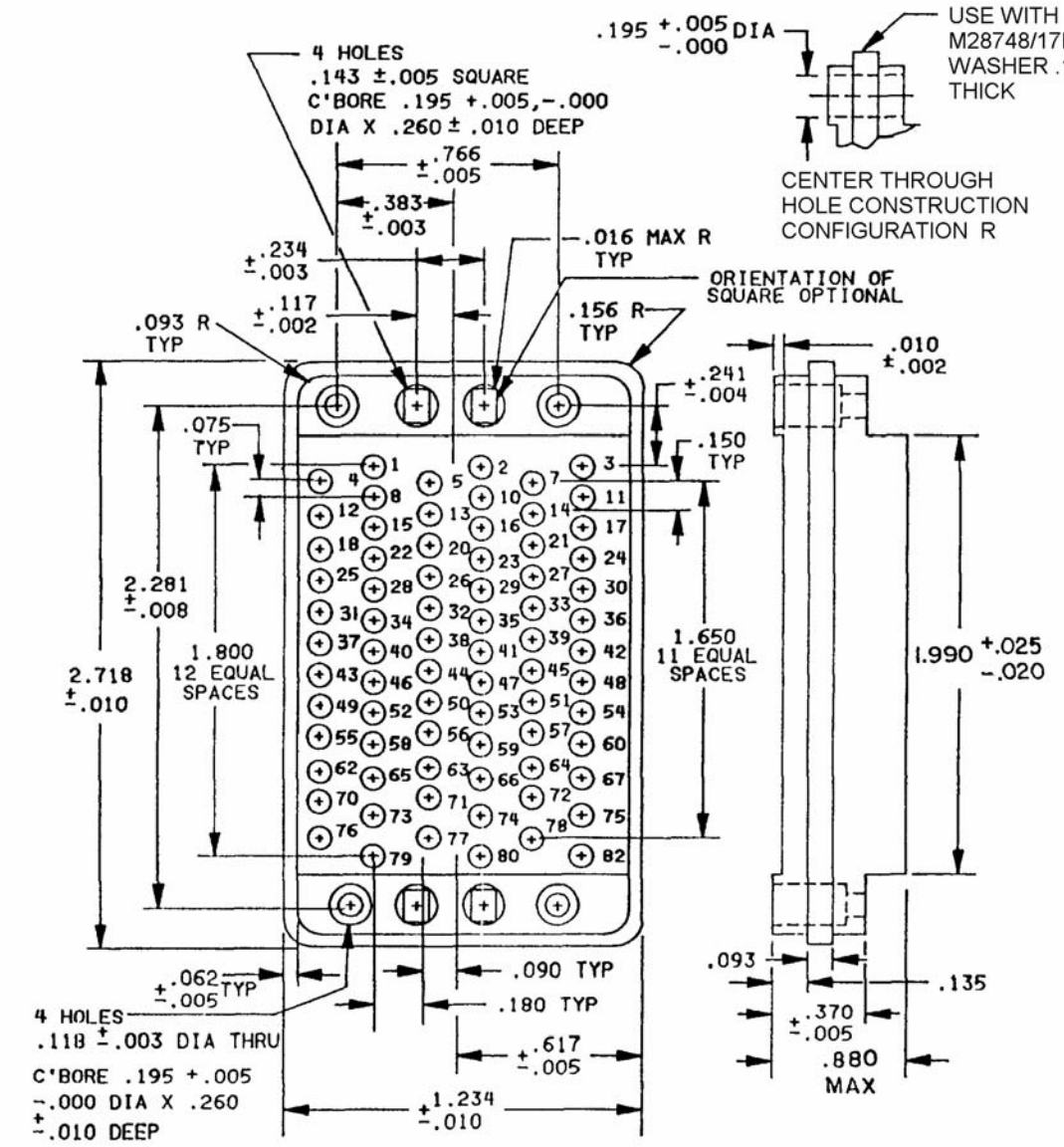
The maximum insertion and withdrawal force (connector assembly) shall be 82.5 lbf (366.98 N).

Insert with 66 contacts

Insert designators J and Q

FIGURE 2. Insert arrangements - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.



Inches	mm
.002	0.05
.003	0.08
.004	0.10
.005	0.13
.008	0.20
.010	0.41
.020	0.51
.025	0.64
.062	1.51
.075	1.91
.090	2.29
.093	2.36
.110	2.79
.117	2.97
.118	3.00
.135	3.43
.143	3.63
.150	3.81
.156	3.96
.180	4.57
.195	4.95
.234	5.94
.241	6.12
.260	6.60
.370	9.40
.383	9.73
.617	15.67
.766	19.46
.880	22.35
1.234	31.34
1.650	41.91
1.800	45.72
1.990	50.55
2.281	57.94
2.718	69.04

Configurations K - Suitable for all applications, except one-piece jackscrews.
Configurations R - For long jackscrews and shield only (through hole alternate construction).

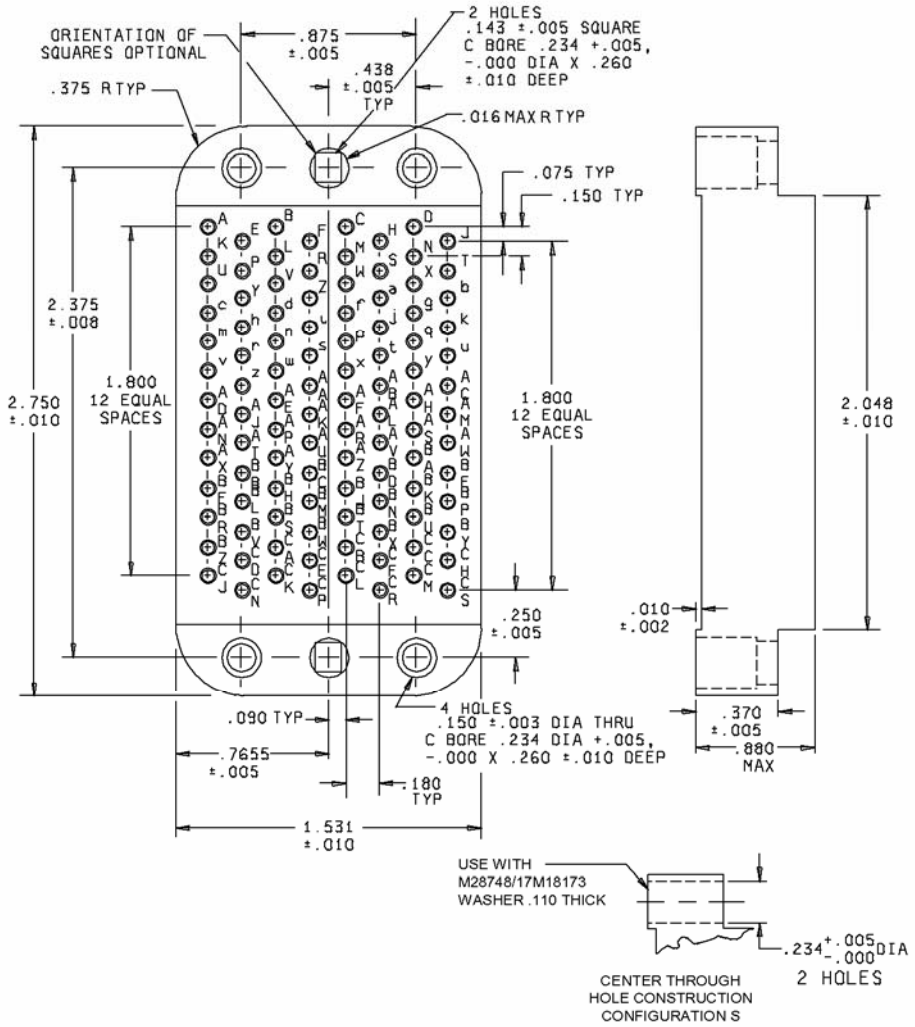
The maximum insertion and withdrawal force (connector assembly) shall be 93.7 lbf (416.80 N).

Insert with 75 contacts

Insert designators K and R

FIGURE 2. Insert arrangements - Continued.

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Inches	mm
.002	0.05
.003	0.08
.005	0.13
.008	0.20
.010	0.25
.016	0.41
.075	1.91
.090	2.79
.110	2.79
.143	3.63
.150	3.81
.180	4.57
.234	5.94
.250	6.35
.260	6.60
.370	9.40
.375	9.53
.438	11.13
.7655	19.444
.875	22.23
.880	22.35
1.531	38.89
1.800	45.72
2.048	52.02
2.375	60.33
2.750	69.85

The maximum insertion and withdrawal force (connector assembly) shall be 130 lbf (578.27 N).

Configuration L - Suitable for all applications, except one-piece jackscrews.
Configuration S - For long jackscrews and shield only (through hole alternate construction).

Insert with 104 contacts

Insert designators L and S

FIGURE 2. Insert arrangements - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figures 1 and 2.

Insert material: See MIL-DTL-28748.

Mating connector: See MIL-DTL-28748/1 and MIL-DTL-28748/3.

Hardware: See MIL-DTL-28748/17.

Part or Identifying Number (PIN): As shown in the following examples. 1/ 2/

See table I for permissible PINs:

(See PIN codes)

M28748/4 –	A	1	H	L	1 A
Basic number of specification sheet	Insert designator	Shield and shield clamp location or retaining plate	Shell and polarization position	Jackscrews or guidepins	Contacts (see table II)

PIN codes:

Insert designator	Number of contact positions
A	7
B	9
C	14
D	20
E	26
F	34
G	42
H	50
J	66
K	75
L	104
Insert designator through hole 6/	
M	34
N	42
P	50
Q	66
R	75
S	104

Shield and shield clamp Location or retaining plate 3/

0 – No shield or retaining plate

Retaining plate

1 – M18198
2 – M18199
3 – M18200

Shield and shield clamp location

4 - top M18192
5-side M18192
6 - top M18193
7-side M18193

Shell and polarization designators 4/

0 - No shell included

Receptacle
A – A of M18204
B – B of M18204
C – C of M18204
D – D of M18204
E – E of M18204
F – F of M18204
G – G of M18204
H - Unpolarized M18204

Plug
J – A of M18203
K – B of M18203
L – C of M18203
M – D of M18203
N – E of M18203
P – F of M18203
Q – G of M18203
R – Unpolarized M18203

Jackscrews or guidepins 5/

0 – No Jackscrews or guidepins.
F - Fixed jackscrews, M18196 (not used with shield).
G - Guidepins, M18197 (not used with shield).
L - Long jackscrews M18194
S - Short jackscrews M18195 (not used with shield).

Jack and guide location: The jackscrew male (turning or fixed) or guide pin shall be located by the A or 1 pin contact.

See footnotes on next page.

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1/ The number zero (0) is used to indicate which parts are not included.
2/ To order individual piece parts for the shield, retaining plate, shell, jackscrews and guide pins, use MIL-DTL-28748/17.
3/ Retaining plate and shells are not available for insert size A.
4/ Polarization positions are for location of plug and receptacle shell pins and slots. See M28748/17.
5/ Jackscrew designator L: Multi-piece jackscrews will be supplied with insert designators A through L. Long one-piece jackscrews will be supplied for insert designators M through S.
6/ Inserts M through S have through holes for use with long one-piece jackscrew designs only.
7/ Mold bodies and hardware from different manufacturers are not necessarily interchangeable.

TABLE I. Permissible PIN's (connector assembly).

<table>
<tr><td>Configuration</td><td colspan="5">Accessories</td></tr>
<tr><td>Receptacle</td><td>Insert</td><td>Shield and clamp position/ or retaining plate</td><td>Shell and polarization position</td><td>Jackscrews</td><td>Contacts</td></tr>
<tr><td rowspan="4">M28748/4-</td><td>A</td><td>0</td><td>0</td><td rowspan="4">0, F, G</td><td rowspan="11">See table II</td></tr>
<tr><td>B
C
D
E</td><td>0
Retaining plate 1</td><td rowspan="3">0,
A thru H</td></tr>
<tr><td>F
G
H
K</td><td>0
Retaining plate 2</td></tr>
<tr><td>J
L</td><td>0
Retaining plate 3</td></tr>
<tr><td>Plug</td><td></td><td></td><td></td><td></td></tr>
<tr><td rowspan="7">M28748/4-</td><td>A</td><td>0</td><td rowspan="3"></td><td rowspan="7">0, L, G, S</td></tr>
<tr><td>B
C
D
E</td><td rowspan="3">0
Shield 4, 5</td></tr>
<tr><td>F
G
H
K</td><td>0</td></tr>
<tr><td>J
L</td><td>0
Shield 6, 7</td><td rowspan="3">0,
J thru R</td></tr>
<tr><td>M
N
P
R</td><td>0
Shield 4, 5</td></tr>
<tr><td>Q
S</td><td>0
Shield 6, 7</td></tr>
</table>

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T14:12Z
Check the source to verify that this is the current version before use.

TABLE II. Contact size percent 1/

Contact	Sizes 16-16 M39029/35-276 (percent)	Sizes 16-20 M39029/35-275 (percent)	Sizes 20-20 M39029/35-274 (percent)
1A	100	0	0
1B	90	0	0
1C	80	0	0
1D	70	0	0
1E	60	0	0
1F	50	0	0
1G	40	0	0
1H	30	0	0
1J	20	0	0
1K	10	0	0
1L	0	0	0
2A	0	100	0
2B	0	90	0
2C	0	80	0
2D	0	70	0
2E	0	60	0
2F	0	50	0
2G	0	40	0
2H	0	30	0
2J	0	20	0
2K	0	10	0
3A	0	0	100
3B	0	0	90
3C	0	0	80
3D	0	0	70
3E	0	0	60
3F	0	0	50
3G	0	0	40
3H	0	0	30
3J	0	0	20
3K	0	0	10

1/ Contact percent size = number of contact positions times the percent in table II. Example: 34 X .80 = 27.2 or 27 contacts supplied. Values of .5 and above are rounded to the next contact.

Crimp tool M22520/1-01.
Positioner M22520/1-03 (positioner A color blue/red).
Installing tool M81969/18-01.
Removal tool M81969/20-01.

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For superseded insert Military Standard sheet PIN's to M28748 PIN's see table III.

TABLE III. Supersession data (inserts).

MS PIN	M28748/4 PIN	Number of contact positions
MS14011-1	M28748/4-A0001L	7
MS14006-1	M28748/4-B0001L	9
MS18175-1	M28748/4-C0001L	14
MS18177-1	M28748/4-D0001L	20
MS14005-1	M28748/4-E0001L	26
MS18179-1	M28748/4-F0001L	34
MS18181-1	M28748/4-G0001L	42
MS18183-1	M28748/4-H0001L	50
MS18185-1	M28748/4-J0001L	66
MS18188-1	M28748/4-K0001L	75
MS18190-1	M28748/4-L0001L	104
MS18178-2	M28748/4-M0001L	34 1/
MS18181-2	M28748/4-N0001L	42 1/
MS18183-2	M28748/4-P0001L	50 1/
MS18185-2	M28748/4-Q0001L	66 1/
MS18188-2	M28748/4-R0001L	75 1/
MS18190-2	M28748/4-S0001L	104 1/

1/ Insert through hole designator. Inserts -2 (M through S) have through holes for use with long one-piece jackscrew designs only.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Reference documents. In addition to MIL-DTL-28748, this document references the following:

MIL-DTL-28748/1
MIL-DTL-28748/3
MIL-DTL-28748/17

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-4752-003)

Review activities:
Army - AT, MI
Navy - AS, SH
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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